HEMANTH VASUDEV N P

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EDUCATION

M.Sc. (Integrated) Software Systems 2024 – 2029
PSG College of Technology, Coimbatore CGPA: 8.43/10

XII (Higher Secondary) 2024
The TVS School, Madurai 90.3%

X (SSLC) 2022
The TVS School, Madurai 88.8%

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL
Frameworks: FastAPI, Flask, Node.js, Express.js, React.js, Spring Boot
Databases: PostgreSQL, MySQL, MongoDB, Redis, Neo4j
Tools: Linux, Docker, GitHub Actions, Git

EXPERIENCE

Backend Engineering Intern | EWhizard Mar 2026 – Aug 2026

- Patched a **cross-user data leak** in LLM-generated SQL on PostgreSQL by forcing every query through a per-user filter, so users of 2 FastAPI endpoints only ever see their own rows.
- Cut categorization API latency **40%** (152 to 92 ms) by profiling a 3-stage ML pipeline served by FastAPI and removing a separate **process spawn** that ran on every prediction.
- Developed a Node.js **MCP server** that lets AI coding agents turn Figma designs into React and CSS, handling a 500-layer screen in **12 ms** and a 12,000-layer page in **104 ms**.

PROJECTS

ArthaDhruva | github.com/hemanthvnp/ArthaDhruva Aug 2026 – Oct 2026

- Architected a **multi-tenant** mortgage risk platform in Spring Boot that scores loans for default via ONNX at **P95 under 50 ms**, isolates tenants with **PostgreSQL RLS** and has 385 tests.
- Engineered a PostgreSQL job queue with lease-based **leader election**, so portfolio runs survive a replica crash and resume from checkpoints. A CI test kills a replica mid-run to prove it.
- Built a regime-switching default model that computes expected loss exactly over all 2^T economic paths in **O(T)** time instead of simulating them, verified against brute-force enumeration.

Throttlr | github.com/hemanthvnp/Throttlr Mar 2026 – Jun 2026

- Implemented a C++20 API gateway with TLS, JWT auth, rate limiting and **circuit-breaker failover**. With a backend killed, **60 of 60** GETs succeed and POSTs run at most once.
- Rewrote the connection layer on an **epoll** event loop so idle clients no longer tie up worker threads, sustaining **over 21K req/s** at 400 connections (P99 under 40 ms).
- Shipped hot reload of routes, rate limits and JWT keys through an **atomic config swap**, with no restart. Under load, 11 reloads produced **zero failed responses** or connection errors.

CineScope | github.com/hemanthvnp/CineScope Feb 2026 – Apr 2026

- Designed a conversational movie search as **4 Dockerized microservices**, running **LiteLLM** intent classification and TMDB prefetch in parallel and streaming results over **SSE**.
- Added a **rule-based fallback** that answers searches when the LLM is down, and kept those degraded answers **out of the cache** so users get full results as soon as it recovers.
- Decoupled model retraining with **Redis Streams** consumer groups, so a burst of ratings triggers one rebuild and events are acked only after it finishes (**at-least-once** delivery).

EXTRACURRICULAR ACTIVITIES

- **CSA Tech Team**: Deputy Coordinator, Computational Sciences Association.
- **FinVerse**: Analyst, Finance Club.